Sales Performance Analysis Dashboard

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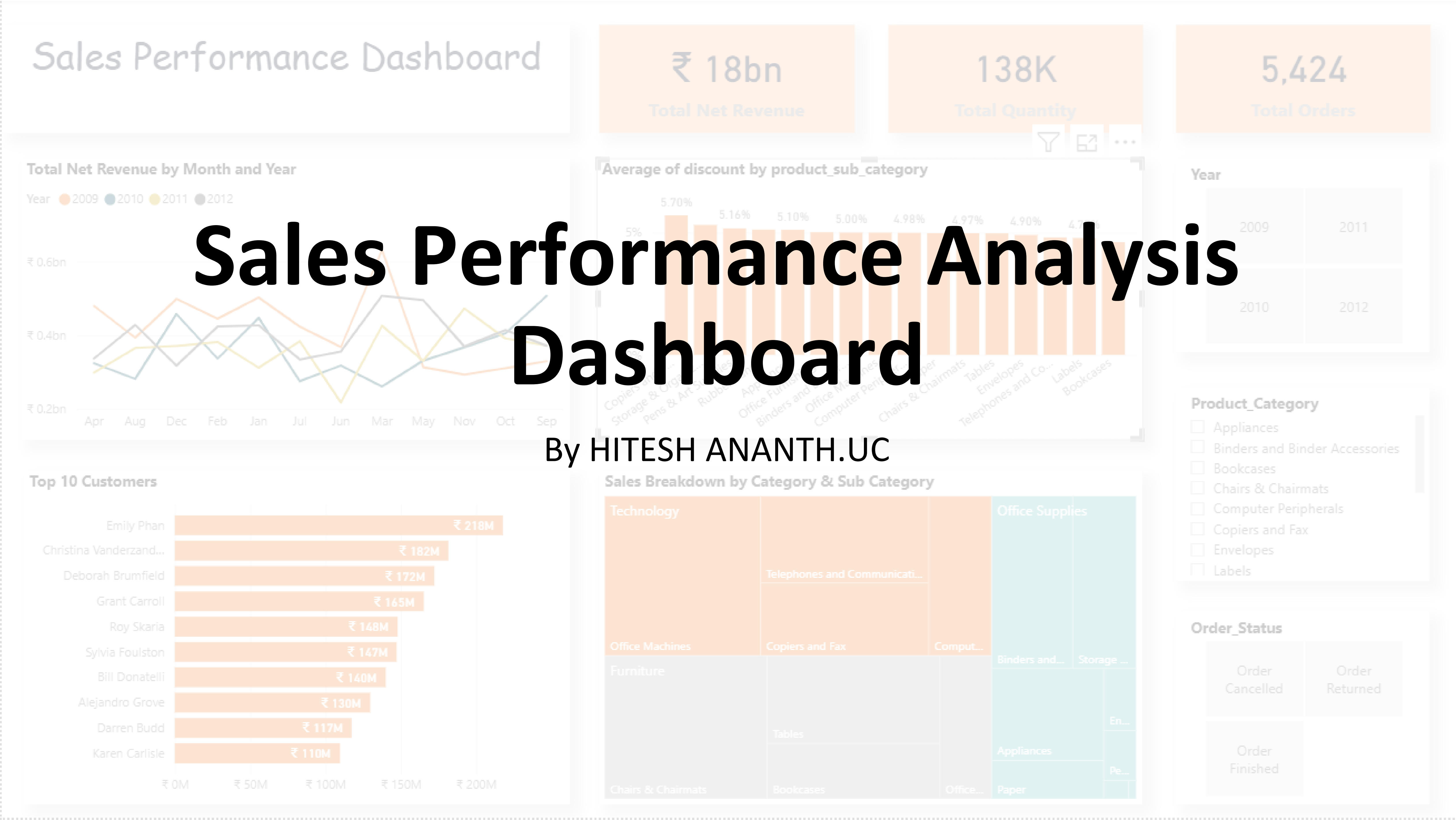


Data Preprocessing – Python

(Transforming Raw Data into a Reliable Dataset)

- **Key Steps in Preprocessing:**
- **Organizing Columns:** Mapped the raw file to 10 specific business columns to ensure every order is tracked correctly.
- **Standardizing Formats:** Converted dates and numbers into formats that Power BI can read without errors.
- **Cleaning Text:** Removed extra spaces and "noise" from the data so categories and customer names match perfectly.
- **Adding Value:** Created the **Net Sales** metric to show actual revenue after discounts are taken out.

```python
import pandas as pd
import numpy as np

# LOAD THE DATA

df_raw = pd.read_csv('raw_data.csv', sep='|', skiprows=2, header=None, engine='python')
```

- **import pandas as pd / import numpy as np**: We load these libraries to perform high-speed data calculations and table manipulations.
- **pd.read_csv('raw_data.csv', ...)**: This line reads the raw file. We use sep='|' because the data is separated by pipes, and skiprows=2 to skip the decorative border lines at the top of the file.

```python
# ORGANIZE COLUMNS

column_names = [
    'order_id', 'order_status', 'customer', 'order_date',
    'order_quantity', 'sales', 'discount', 'discount_value',
    'product_category', 'product_sub_category'
]
df = df_raw.iloc[:, 1:11]
df.columns = column_names
```

- **column_names** = [...]: We manually define the headers to ensure the data is labeled correctly (ID, Sales, Discount, etc.).

- **df = df_raw.iloc[:, 1:11]:** Since the pipe format created empty columns at the edges, this line selects only the "inner" 10 columns containing our actual data.

- **df.columns** = column_names: We officially apply our clean headers to the table.

```python
#  CLEAN TEXT & DATES

for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = df[col].str.strip()

#  format
df['order_date'] = pd.to_datetime(df['order_date'], errors='coerce')
```

- **for col in df.columns... str.strip():** This loop looks at every text column and removes extra spaces. This ensures that " Furniture" and "Furniture" are treated as the same category.
- **pd.to_datetime(..., errors='coerce'):** This is a critical line. It converts "text dates" into actual Calendar Dates. The errors='coerce' part ensures that if there is a broken date, it doesn't crash our script; it simply marks it as empty.

```python
#  CLEAN NUMERICAL DATA
numeric_cols = ['order_id', 'order_quantity', 'sales', 'discount', 'discount_value']
for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')

# Create Net Sales (Revenue after discounts)
df['net_sales'] = df['sales'] - df['discount_value']

# Remove any empty rows
df = df.dropna(subset=['order_id'])
```

- **pd.to_numeric(..., errors='coerce'):** We force columns like 'Sales' and 'Discount' to be numbers. This prevents Power BI from thinking a price is just a piece of text.
- **df['net_sales'] = df['sales'] - df['discount_value']:** This is our Business Logic. We calculate the true revenue by subtracting the discount given from the total sales price.
- **df.dropna(subset=['order_id']):** This line deletes any completely blank rows that may have been at the bottom of the raw file, ensuring our total order count is 100% accurate.

```python
#  EXPORT
df.to_csv('cleaned_sales_data.csv', index=False)

print("✅ Data cleaning complete!")
print(f"Total rows processed: {len(df)}")
df.head()
```

- **df.to_csv('cleaned_sales_data.csv'):** We save our hard work into a new, clean file. This is the file we actually connect to Power BI.

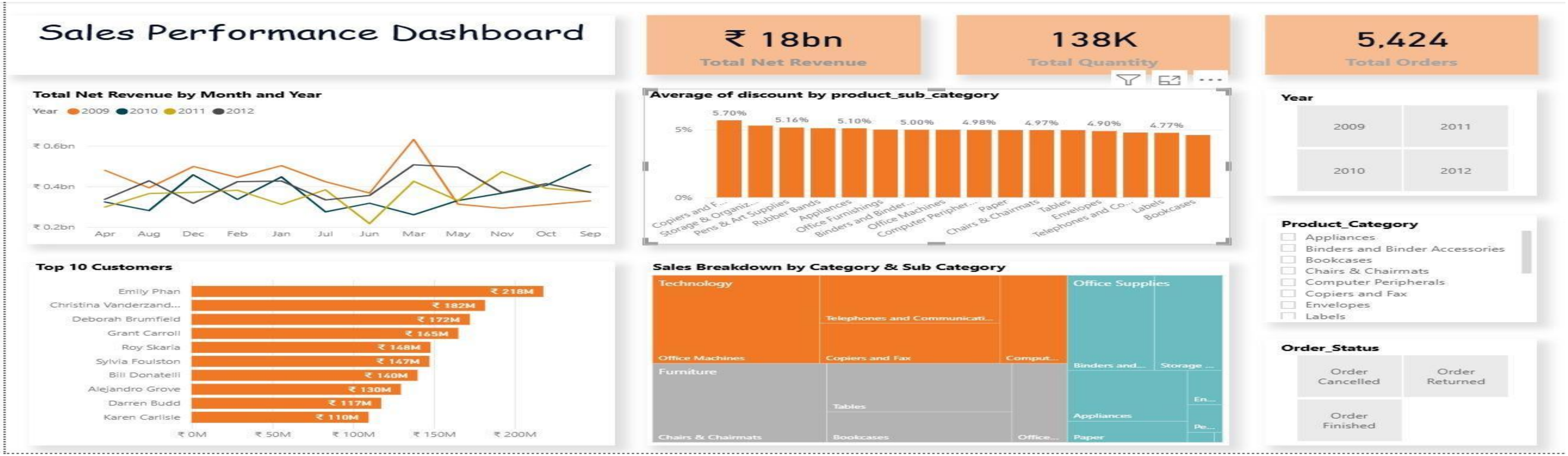
- **print(...):** Provides an instant confirmation in the console that the cleaning was successful.

- **df.head():** This displays the first 5 rows of the final data so we can do a "manual spot check" to verify everything looks perfect before moving to the dashboard.

Dataset Overview

This slide defines what the data actually represents.

- **Dataset Name:** DQLab Store Sales Performance.
- **Timeframe:** 4 Years of Historical Data ($2009 - 2012$).
- **Scale of Operations:**Total Revenue: ₹18 Billion (The financial footprint).
- **Total Orders:** $5,421$ (The number of successful transactions).
- **Total Quantity:** $138,227$ (The volume of products shipped).
- **Data Scope:** Covers 3 major Categories (Technology, Furniture, Office Supplies.



Key Performance Indicators

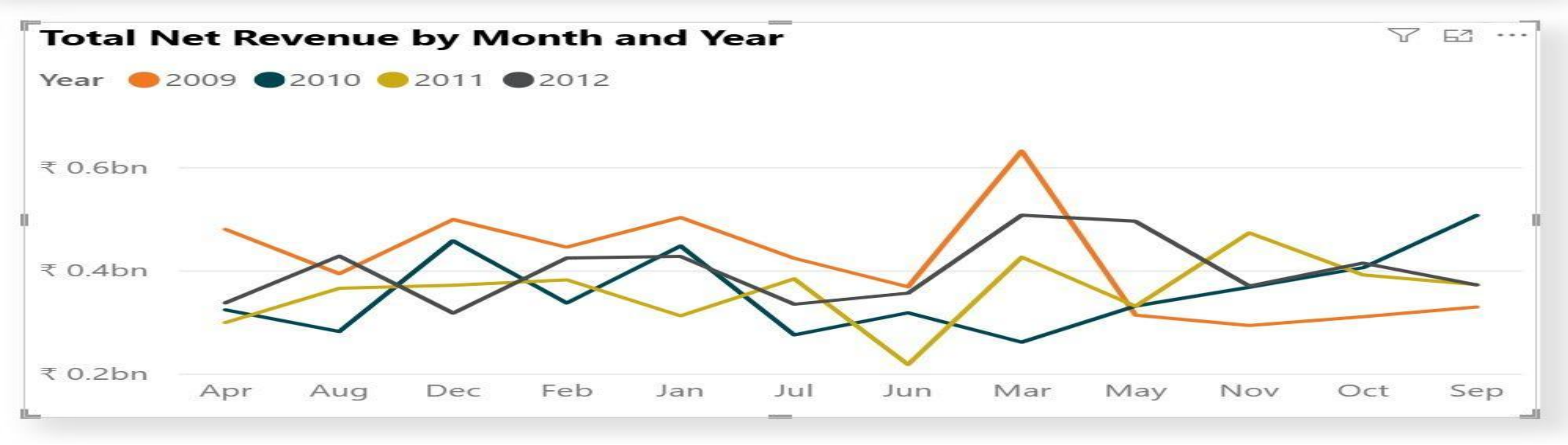
What I did:

- **Metric Aggregation**: I created three "Card" visuals to summarize the entire dataset.

- **Currency Formatting**: Applied Indian Rupee formatting to the ₹18bn Revenue so it aligns with our local business context.

- **Visual Priority**: I placed these at the very top of the dashboard. In Data Analytics, we call this the "High-Value Zone"—it
- ensures a manager sees the most important numbers in the first 2 seconds.

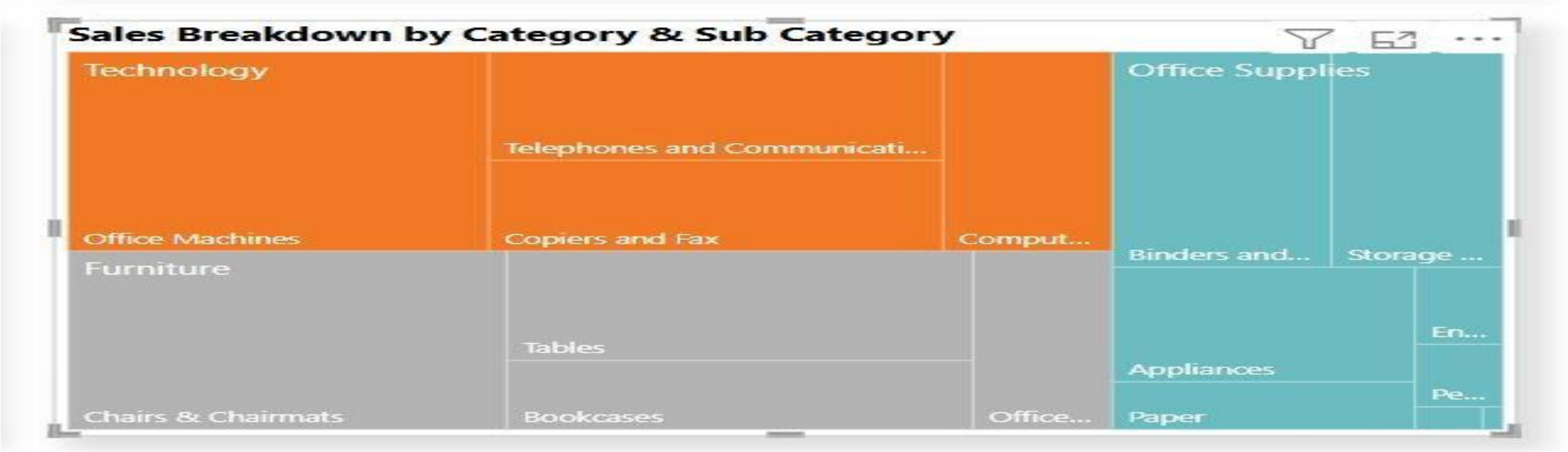
Revenue Trend Analysis



What I Did in this Visual:

- **Metric Mapping**: I plotted Total Net Sales on the vertical axis (Y-axis) against the Order Date on the horizontal axis (X-axis).
- **Time-Series Tracking**: I configured the chart to show the revenue flow over the 4-year period ($2009–2012$) to identify business growth patterns.
- **Visual Clarity:** I used a clean line format to highlight the "peaks" and "valleys" in our sales cycle, making it easy to spot which years performed best.

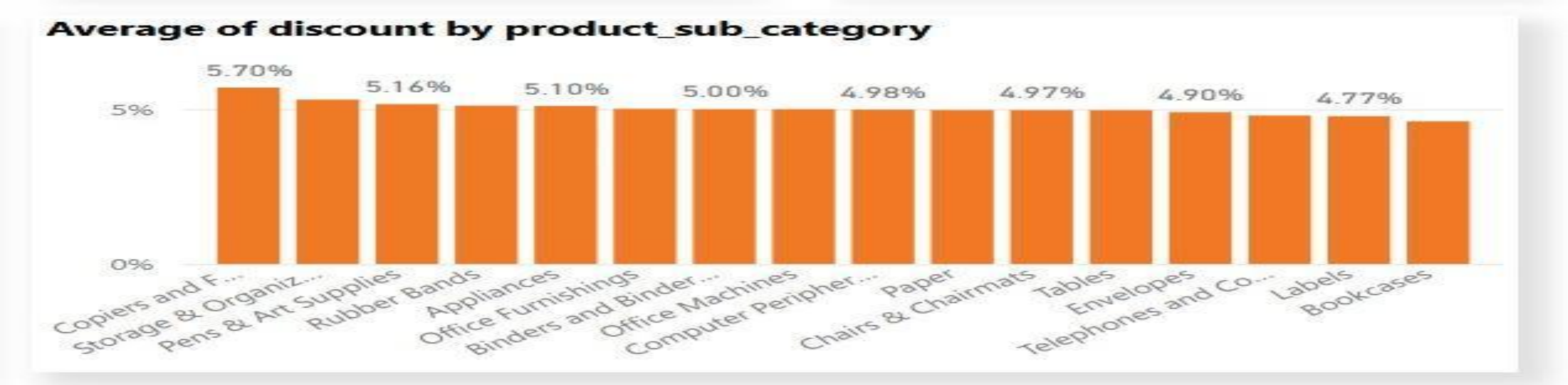
Category Market Share



What I did:

- **Market Share Visualization:** Instead of a basic Pie Chart, I used a Treemap. It’s much better for comparing the size of categories like Technology vs. Furniture.
- **Color Coding**: I used distinct colors for each category to make the distribution of our ₹18bn revenue easy to distinguish at a glance.
- **Insight:** This shows that Technology is our primary revenue driver, occupying the largest square on the map.

Discount Analysis



What I did:

- **Calculated Averages:** Instead of showing the total discount, I changed the value to Average Discount. This is a more accurate way to measure our pricing strategy.

- **Ranking:** I sorted the bars in descending order so the manager can instantly see which products have the highest "price leakage."

- **Insight**: It highlights that Copiers and Fax are being discounted at 5.7%—a key point for the manager to review.

Top 10 Customers

What I did:

- **Visual Choice**: A Bar Chart to create a leaderboard.
- **Filtering**: I applied a 'Top N' filter in the filter pane to show only the Top 10 customers by revenue.
- **The Insight** :
- "Our revenue is driven by high-value individuals. Emily Phan is our leading client. Identifying these top 10 names allows the sales team to focus on 'Account Management' for our most valuable partners."

Interactive Slicers

What I did:

- **The "Control Center":** I added Slicers for Year, Product Category, and Order Status.
- **Dynamic Filtering:** I linked these slicers to every other visual on the page.
- **Practical Use:** "Sir, if you want to see the performance for only 2012 or only Finished Orders, you just click these buttons, and the entire dashboard updates in real-time."

Project Conclusion & Key Takeaways

Technical Success: The "Single Source of Truth"

- **Data Reliability:** By using Python for Preprocessing, we eliminated manual entry errors and "noise" from the raw files.
- **Accuracy:** The ₹18 Billion revenue reported is a validated "Net" figure, accounting for all discounts, providing a reliable foundation for financial decisions.

Strategic Business Insights

- **The "March Peak":** Our data proves a consistent sales surge in March. We should ensure maximum inventory levels and staff availability during this window.
- **The "June Gap":** The trend analysis identified a recurring mid-year dip. This presents a clear opportunity for "Mid-Year Clearance" sales to stabilize revenue.
- **High-Value Segments:** Technology is our primary growth engine, while Copiers are our most heavily discounted items. Reviewing the discount strategy for Copiers could directly increase our profit margins.

Final Value :

Interactivity: Unlike a static Excel report, this Power BI dashboard allows management to filter by Year or Order Status to get answers in seconds rather than hours.